

CHAPTER 1

INTRODUCTION

1.1 Background

Electricity plays a vital role in everyday life, powering homes, industries, and modern electronic devices. However, traditional energy meters only display the total units consumed and do not provide any real-time information about voltage, current, power usage, or daily consumption patterns. Because of this limitation, users often remain unaware of their energy usage until they receive their electricity bill. This lack of monitoring leads to unnecessary power consumption, high electricity bills, and inefficient energy management.

With rapid advancements in the Internet of Things (IoT), it is now possible to measure, record, and access electrical parameters remotely. IoT-based smart meters provide real-time monitoring, daily logs, cost calculations, and remote load control, making energy management far more efficient and user-friendly.

1.2 Need for Smart Energy Monitoring

The growing demand for energy and the rising cost of electricity have increased the need for smart monitoring systems. Users require continuous visibility of their consumption so they can reduce wastage and improve efficiency. Smart meters offer several advantages over conventional meters, such as:

- Real-time tracking of voltage, current, and power
 - Accurate calculation of energy usage (kWh)
 - Automatic logging of daily and monthly data
 - Cost estimation based on consumption
 - Remote ON/OFF control of appliances
- These features help consumers monitor their usage patterns and adopt energy-saving habits.

1.3 Overview of the Proposed System

This project proposes an IoT-Based Smart Energy Meter using an ESP32 microcontroller connected to a voltage sensor, current sensor, and relay module. The ESP32 continuously measures electrical parameters and transmits the data to the Blynk Cloud platform via Wi-Fi. Users can view real-time readings, energy usage graphs, and cost calculations through the Blynk mobile app. The integrated relay allows remote control of electrical loads, adding convenience and safety to the system.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Review of Existing Research Work

Paper 1: Smart Energy Monitoring System Using ESP32 – e-Prime Journal (2024)

El-Khozondar et al. (2024) developed an IoT-based smart energy monitoring system using the ESP32 to measure voltage, current, power, and energy in real time. The data is transmitted to the Blynk app and WhatsApp for user notification, providing remote visibility of energy usage. However, the system does not include relay-based load control or a local LCD display, limiting direct appliance interaction. [1]

Paper 2: ESP32-Based Electric Energy Consumption Meter – Macheso & Thotho (2022)

Macheso and Thotho (2022) developed an ESP32-based IoT energy meter using the SCT-013 current transformer and ZMPT101B voltage sensor to measure RMS voltage, current, active power, and total energy consumption. The system uploads real-time data to the Blynk cloud for remote monitoring and demonstrates good measurement accuracy. However, the design does not include relay-based load switching or advanced control features. [2]

Paper 3: IoT-Based Smart Energy Meter Using ESP32 – ICCIP 2021 (Pimple et al., 2021)

Pimple et al. (2021) developed an IoT-based smart energy meter using the ESP32 microcontroller and the HLW8012 energy sensing chip to measure voltage, current, power, and energy. The system sends consumption data to the user and electricity provider through the Virtuino mobile app using a cloud-connected ESP32, eliminating manual meter reading. While the system successfully supports automatic billing and remote monitoring, it does not implement relay-based load control or cost-saving automation. [4]

Paper 4: Low-Cost IoT-Based Smart Home Automation System – Addow et al. (2025)

Addow et al. (2025) presented a smart home automation system using the ESP32 microcontroller to control appliances, monitor security, and manage energy usage in urban households. The system integrates sensors, smart relays, an ESP32-CAM module, and a mobile app to provide real-time control and monitoring. While the system improves energy efficiency and security, it does not focus specifically on voltage and current measurement or power metering, making it less suitable as a dedicated energy meter. [5]

2.2 Problem Statement

- Existing smart energy meters provide only partial functionality because many designs measure only voltage or only current, resulting in incomplete monitoring.
- Some systems offer IoT connectivity but lack features like energy (kWh) calculation, cost estimation, and local LCD display, and several solutions do not include a relay module for remote load control.
- Many existing approaches require complex servers, making them unsuitable for simple household use.

- The lack of a unified system that combines sensing, monitoring, cost estimation, and load control prevents users from effectively managing their electricity usage, leading to poor energy efficiency and higher operational costs.

2.3 Research Gap

Even though various IoT-based energy meters exist, several gaps remain:

- Many systems measure only voltage or only current.
- Most do not include a relay for remote ON/OFF control.
- Cost estimation and billing features are missing.
- Complex servers are used instead of simple mobile dashboards.
- Local display features such as LCD screens are often not included.
- Few systems integrate all features into a single compact design.

2.4 Objectives

The objectives of this project are:

1. To develop a compact and low-cost IoT-based smart energy meter using ESP32 capable of measuring AC voltage, current, power, and energy (kWh) with sensors such as ZMPT101B and ACS712.
2. To provide real-time monitoring and visualization of electrical parameters through a 16×2 I2C LCD for local display and the Blynk cloud platform for remote access.
3. To enable remote ON/OFF control of household electrical loads using a relay module and support energy tracking to help users reduce power consumption.

CHAPTER 3

METHODOLOGY OF LITERATURE

3.1 Identification of Relevant Studies

Research papers, journals, project reports, and technical articles related to IoT, smart meters, ESP32, current sensors, voltage sensors, and cloud dashboards were selected. Sources included:

- IEEE papers
- ResearchGate publications
- IoT project case studies

3.2 Comparison of Technologies

The following factors were compared across the literature:

- Type of microcontroller used
- Voltage and current measurement techniques
- IoT communication methods (Wi-Fi, MQTT, HTTP)
- Cloud dashboard features
- Accuracy, cost, and reliability
- Inclusion of relay control
- Energy (kWh) and cost calculation methods

3.3 Selection Criteria

Studies were included based on:

- Relevance to smart energy monitoring
- Practical implementation
- Use of modern IoT platforms
- Accuracy of sensing components
- Real-time data logging capability

This approach helped identify the most efficient components and techniques for building the proposed system.

3.4 Component Details from Literature

Component analysis is a major part of methodology because each sensor contributes directly to the accuracy and performance of the smart energy meter. The following components are widely referenced in previous research and also used in the proposed system.

3.4.1 ESP32 Microcontroller



The ESP32 is a powerful, low-cost microcontroller widely used in IoT applications due to its built-in Wi-Fi and Bluetooth capabilities. It features a dual-core processor with high clock speed, multiple analog-to-digital converter (ADC) pins, and excellent power efficiency. These features make it ideal for collecting sensor data, performing real-time calculations, and sending information to cloud platforms such as Blynk. Compared to older boards like Arduino, the ESP32 offers faster processing, greater memory, and seamless wireless connectivity, making it highly suitable for smart energy monitoring systems.

Figure 3.4.1

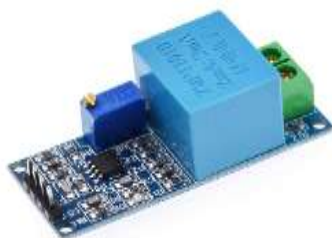
3.4.2 ACS712 Current Sensor



The ACS712 is a Hall-effect based current sensor used for measuring AC or DC current with good accuracy and stability. It outputs an analog voltage that varies proportionally with the current flowing through the conductor. The sensor is easy to interface with microcontrollers and provides isolation between the high-voltage load and the ESP32, ensuring safety. Its low noise and sensitivity make it ideal for real-time current monitoring in smart meter applications. Different variants (5A, 20A, 30A) are available, allowing it to be used with various load

Figure 3.4.2

3.4.3 ZMPT101B Voltage Sensor



The ZMPT101B is a precise AC voltage sensor designed for safe and accurate measurement of mains voltage. It uses a small voltage transformer to isolate high voltage from the microcontroller, making it safe for electronics and users. The sensor provides a clean analog output that can be calibrated to measure real-time voltage levels. Due to its stability, isolation, and ease of interfacing with ADC pins, the ZMPT101B is widely used in energy monitoring systems, power analyzers, and IoT-based smart meters.

Figure 3.4.3

3.4.4 Relay Module (5V, 1-Channel)



Figure 3.4.4

The 5V single-channel relay module is used to control AC appliances through the microcontroller. It acts as an electrically operated switch that allows the ESP32 to turn loads ON or OFF remotely. The relay provides complete isolation between the ESP32's low-voltage control circuitry and the high-voltage AC load, ensuring safety and preventing damage to electronic components. This makes it ideal for applications such as smart home automation, remote appliance control, and load switching within smart energy

3.4.5 LCD 16×2 with I2C Module



Figure 3.4.5

The 16×2 LCD display with I2C interface provides a simple and efficient way to show real-time readings such as voltage, current, power, and energy consumption. The I2C module reduces the number of required connection pins from 16 to only 4, making wiring easier and more reliable. The display is clear, low-power, and programmable, making it perfect for monitoring systems where live data needs to be shown locally in addition to being sent to the cloud. Its compatibility with the ESP32 and availability of ready-made

3.5 Block Diagram

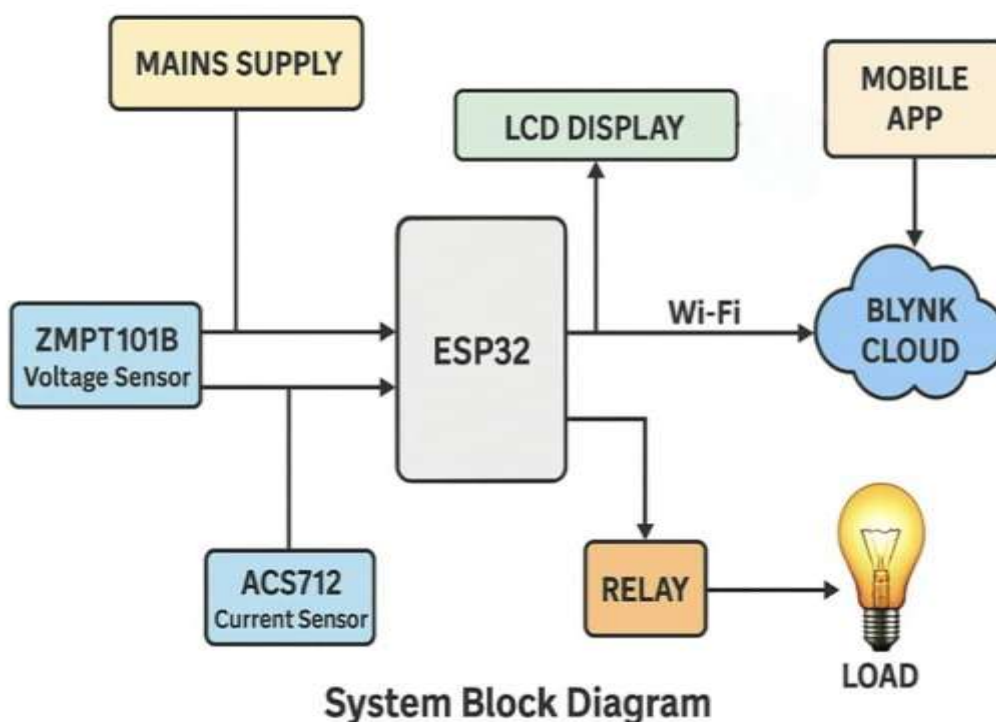


Figure 3.5.1 BLOCK DIAGRAM

3.5.1 Block Diagram Explanation

The block diagram represents the complete hardware architecture of the IoT-Based Smart Energy Meter using the ESP32 microcontroller. All sensors and modules are interfaced to the ESP32 through its GPIO pins, making the system efficient, modular, and easy to integrate.

The ZMPT101B voltage sensor measures the AC voltage from the mains supply. Its analog output is connected to GPIO34 (ADC1_CH6) of the ESP32. Likewise, the ACS712 current sensor measures current drawn by the load and sends its analog output to GPIO35 (ADC1_CH7). These ADC-enabled GPIO pins allow the ESP32 to accurately read continuous analog signals from both sensors.

The 16×2 LCD with I2C interface is connected using only two GPIO pins: GPIO21 (SDA) and GPIO22 (SCL). This significantly reduces wiring complexity compared to traditional parallel LCD connections and allows clear real-time display of voltage, current, power, and energy readings.

A 5V relay module is connected to an output-capable GPIO pin (in this design, GPIO22 or any suitable digital GPIO). This relay allows the ESP32 to switch the connected AC load ON or OFF remotely through commands received from the Blynk mobile app. This ensures complete monitoring and control of the electrical load.

All modules—including sensors, relay, LCD, and ESP32—share a common GND, ensuring stable reference voltage and noise-free ADC readings. Power to the ESP32 is supplied through its VIN (5V) pin, and the sensors and relay are powered appropriately from the same supply.

Once the ESP32 reads and processes the sensor values, it transmits all real-time data to the Blynk Cloud over Wi-Fi. The user can monitor these values and also control the relay using the Blynk mobile application, completing the IoT integration.

This block diagram therefore clearly shows the path of power, the flow of data from sensors to ESP32 through GPIO pins, the output control to the relay, and the communication link to the cloud and mobile app.

3.6 Flowchart

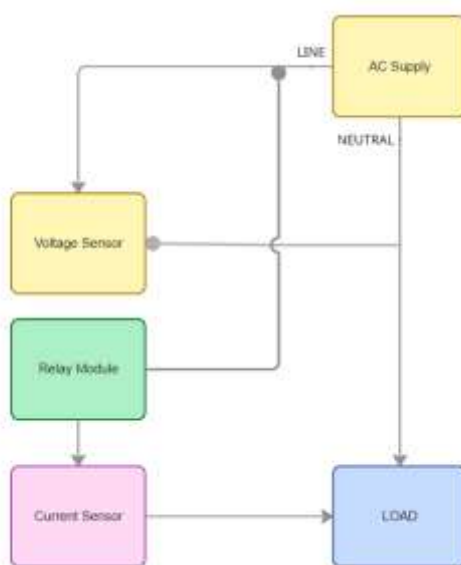


Figure 3.6.1 FLOW OF CURRENT

3.6.1 Flowchart Explanation

This flowchart illustrates the flow of electrical power and the sensing process in the Smart Energy Meter system. The AC supply provides two wires: Line and Neutral. The Neutral wire goes directly to the load without any interruptions, ensuring a stable reference for voltage and current measurement. The Line wire is routed through different components to enable sensing and control.

First, a branch from the Line wire is connected to the Voltage Sensor, allowing the system to measure the real-time AC voltage supplied to the load. The main Line then continues through the Relay Module, which functions as an electronically controlled switch. This enables the ESP32 to turn the load ON or OFF remotely using IoT commands sent from the Blynk application.

After the relay, the Line wire passes through the Current Sensor, which measures the current drawn by the connected load. Placing the current sensor in series ensures precise current sensing under all operating conditions.

This arrangement ensures:

- Accurate and isolated voltage measurement
- Precise and stable current sensing
- Safe and effective load control through the relay
- A clean measurement path with minimal electrical interference

The flowchart clearly demonstrates how the electrical parameters are sensed and how the relay enables remote load control while keeping the Neutral line uninterrupted. This design ensures both electrical safety and accurate measurement for the Smart Energy Meter.

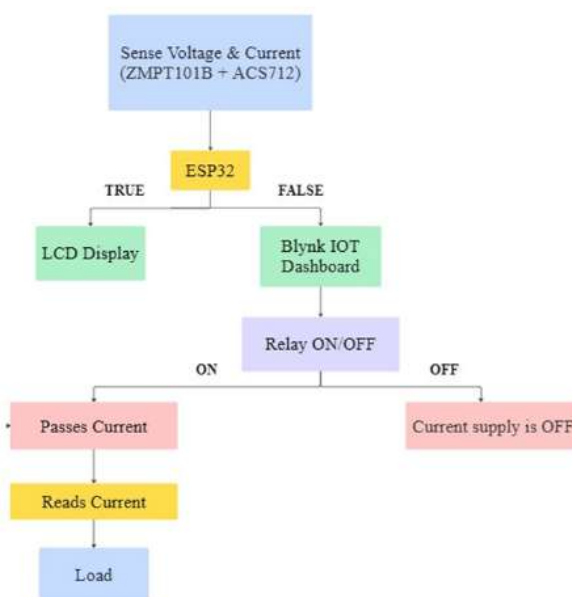


Figure 3.6.2 FLOWCHART

3.6.2 Flowchart Explanation

This flowchart represents the decision-making process of the Smart Energy Meter once voltage and current values are sensed by the ZMPT101B and ACS712 sensors. The ESP32 receives the sensor data and determines how the information should be processed and displayed.

After sensing the electrical parameters, the ESP32 evaluates the availability of a local or remote display path. If the local LCD display is active (TRUE condition), the ESP32 sends the measured voltage, current, and power values to the LCD, enabling the user to view the readings directly on the device. If local display is not active (FALSE condition), the ESP32 transmits the values to the Blynk IoT Dashboard for remote monitoring.

Next, the flowchart shows the decision process for controlling the relay module. The relay state (ON/OFF) is determined by the user's input from the Blynk application or predefined system logic.

- When the relay is ON, current flows to the load. The current sensor reads the load current, and the ESP32 updates the system with real-time consumption data.
- When the relay is OFF, the current supply to the load is cut off, ensuring that no electrical power reaches the connected device.

This flowchart clearly illustrates how the system handles sensor inputs, chooses the correct display path, and manages load control using the relay. It demonstrates the logical sequence that ensures safe, accurate, and reliable operation of the Smart Energy Meter under both local and remote monitoring conditions.

CHAPTER 4

Results and Discussions

4.1 Visual Results

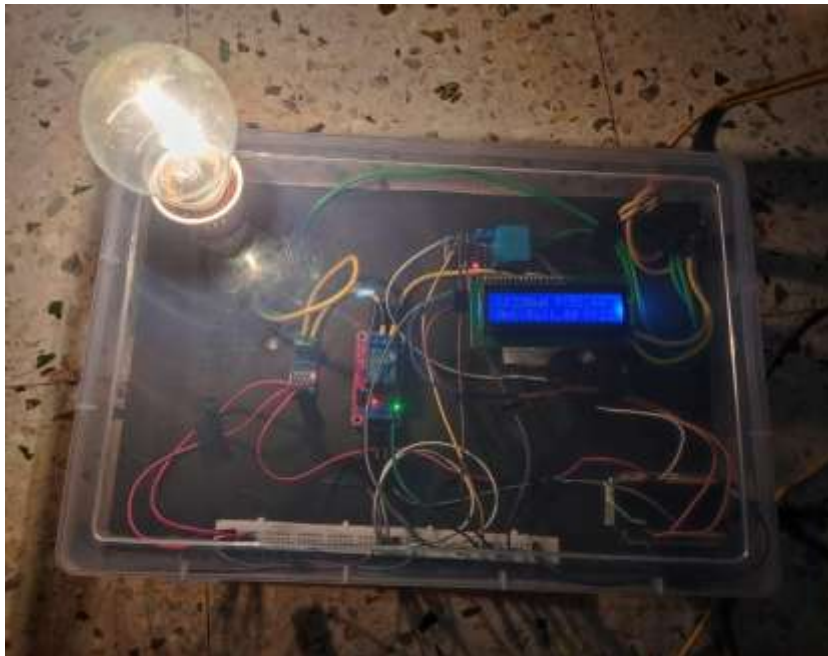


Figure 4.1.1 HARDWARE SETUP

The assembled prototype of the IoT-based Smart Energy Meter is shown in Fig.4.1. The entire system is mounted inside a transparent enclosure for safety and clear visibility of the circuitry. The ESP32 microcontroller is positioned at the center, acting as the main controller for sensing, processing, and communication. The ZMPT101B voltage sensor and ACS712 current sensor are connected to the AC line and are placed on the left side of the setup to measure real-time voltage and current, respectively. A 16×2 I2C LCD is mounted at the top-right, displaying live system parameters such as voltage, current, power, energy consumption, and cost. The relay module is connected in series with the load to enable remote ON/OFF control through the Blynk IoT platform. A light bulb is used as the test load, and its illumination confirms successful relay operation and power delivery. Proper wiring, grounding, and insulation have been maintained to ensure safe operation. This hardware setup demonstrates the complete working of the smart energy meter, integrating sensing, processing, display, and IoT-based control in a compact and functional prototype.

Observations from the Dashboard:



Figure 4.1.2 BLYNK IOT DASHBOARD IN DESKTOP



Figure 4.1.3 BLYNK IOT DASHBOARD FOR MOBILE

The Blynk IoT dashboard displays the real-time electrical parameters measured by the smart energy meter. The voltage, current, power, and energy (kWh) values are continuously updated from the ESP32 through Wi-Fi. Each gauge widget shows the live readings, helping users easily monitor their electricity usage. The dashboard also includes a switch widget that allows remote control of the connected load through the relay module. Additionally, an amount widget is used to estimate the electricity cost based on the energy consumed. Overall, the dashboard provides a simple and user-friendly interface for monitoring and controlling the smart energy meter from anywhere.

CHAPTER 5

Conclusion And Future scope

5.1 Conclusion

The IoT-Based Smart Energy Meter developed in this project accurately measures voltage, current, power, and energy using the ESP32, ZMPT101B, and ACS712 sensors. Real-time values are displayed on a 16×2 I2C LCD, while the Blynk IoT platform enables remote monitoring and relay-based load control through a mobile app. The system demonstrated stable sensor readings, fast data transmission, and reliable cloud connectivity after calibration. Overall, the project achieves its objectives by providing a low-cost, efficient, and user-friendly smart energy meter suitable for home and small commercial applications.

5.2 Future Scope

1. Automated Billing and Reports:

Future versions may include automatic bill generation based on recorded energy usage. The system can also provide monthly or weekly reports, helping users clearly understand their consumption patterns and manage electricity usage more effectively.

2. Enhanced Data Security and Cloud Storage:

The smart meter can be upgraded with encrypted data transfer and secure authentication. Cloud storage can also be added to store long-term consumption data, allowing users to view their past usage and access records anytime.

3. AI-Based Load Prediction and Optimization:

Artificial intelligence can be integrated to predict future power consumption patterns and suggest optimized usage. This helps users reduce unnecessary energy wastage and minimize electricity bills by managing loads efficiently.

4. Industrial-Grade Upgrades:

The system can be improved for industrial applications by using higher-accuracy sensors that provide precise measurements even at large loads. Additionally, the overall design can be made more robust to withstand heavy electrical loads and harsh environmental conditions, making the smart meter reliable for factory and commercial use.

References

- [1] S. El-Khozondar *et al.*, “A Smart Energy Monitoring System Using ESP32 Microcontroller,” *Journal of Sustainable Energy Systems*, 2024.
- [2] F. Macheso and L. Thotho, “ESP32 Based Electric Energy Consumption Meter,” *International Journal of Computing and Communication Innovation*, 2022.
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